
FORMATION OF MONOPOLIES IN THE DIGITAL ECONOMY

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- How digital platforms become dominant
 - Why some companies become impossible to replace
 - The forces that create “winner-take-most” markets

Examples

- Google Search → 90% global market share
- Meta → owns Facebook + Instagram + WhatsApp
- Apple → highest consumer lock-in ecosystem

WHY STUDY DIGITAL MONOPOLIES?

- Digital markets produce a few “big winners”
 - Competition behaves differently than traditional markets
 - Must understand *why giants stay giants*
 - Predict market movement
 - See why startups fail or succeed
 - Examples:
 - Google Search never got replaced after early 2000s
 - TikTok disrupted video space but still faces platform dominance rules
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WHAT IS A MONOPOLY IN DIGITAL ECONOMY?

- Not always “100% market control”
 - Often “winner-take-most”
 - Dominance based on:
 - Network effects
 - Control over data
 - High switching costs
 - Platform ecosystem scale
 - WhatsApp in Nepal/India — nearly universal messaging
 - Google Maps — default navigation globally
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Market type	Examples
De facto monopoly: There is only one seller/provider of the goods/services and no substitutes.	Amazon is the world's largest e-commerce site, Meta controls social networking, and Microsoft dominates business software.
Oligopoly: There are only a few sellers of the same products/services.	Apple and Samsung lead the smartphone market.
Market with monopolistic competition: There are many sellers in the same marketplace offering differentiated products/services.	Fast food chains like Subway , McDonald's , KFC , Pizza Hut , Burger King and Domino's .
Monopsony: There is only one buyer.	Amazon is the most prominent buyer/distributor in the e-book market. YouTube is the main channel for producers to distribute video clips, though TikTok rules the youth market.
Oligopsony: There are more than one (but just a few) buyers.	Google , Facebook , and Amazon function as dominant buyers of digital advertising space, not least because of their unparalleled access to user data.

METRICS USED TO IDENTIFY MONOPOLY

- The company is a major player whose products account for a significant share of users or traffic.
- The company's products have no competitors, or there is little competition on the market.
- The company obtains significant competitive advantages in related products due to its dominant position in one market.

WHY DIGITAL MARKETS TEND TOWARD MONOPOLIES

- Network effects → more users = more value
 - Low cost to scale → almost free after setup
 - Lock-in → users don't want to switch
 - Data advantage → impossible for competitors to copy
 - Platform bundling → everything inside one ecosystem
 - Examples
 - Apple ecosystem (iPhone + iCloud + AirPods) locks users in
 - Amazon's fast shipping → attracts sellers → attracts buyers
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ECONOMIES OF SCALE (DEFINITION)

- **“Cost per user decreases as platform gets bigger.”**
 - High fixed cost → low incremental cost
 - Bigger platforms operate cheaper & faster
 - Small competitors cannot match efficiency
 - Examples
 - Netflix producing a show once → millions watch
 - Amazon warehouses → cost per order becomes cheaper as volume grows
 - Google Search servers → same cost whether 1M or 100M users
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ECONOMIES OF SCALE (EXAMPLES)

- Facebook hosting a new user costs almost nothing
- Microsoft's cloud servers (Azure) cost less per customer as scale increases
- Netflix : Produces a show once → millions of viewers stream it → cost per viewer is tiny.
- Apple producing millions of iPhones reduces manufacturing cost/unit

Real fact

- Amazon's shipping cost per package decreased every year as volume increased.
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ECONOMIES OF SCOPE (DEFINITION)

- **“Use same infrastructure to offer more services.”**
 - One login → many apps
 - One dataset → multiple products
 - One user → multiple monetization streams
 - Examples
 - Google Account → Gmail + Maps + YouTube + Drive
 - Apple ID → iPhone + Mac + iCloud + Apple Music
 - Amazon → Ecommerce + AWS + Prime Video + Logistics
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ECONOMIES OF SCOPE (EXAMPLES)

- Uber uses same driver network for UberX, UberMoto, Uber Eats
- Meta uses the same ad network across Facebook, Instagram, WhatsApp
- ByteDance uses one algorithm foundation for TikTok + CapCut
- Like a swiss army knife, one tool (platform) – multiple usage (services)



DATA ADVANTAGE (DEFINITION)

- **“More users = more data = better product = more users.”**
 - Data improves personalization
 - Competitors cannot easily replicate years of data
 - AI models improve with user input
 - Examples
 - TikTok’s For You Page learns from billions of interactions
 - Google Maps traffic predictions → based on real-time movement
 - Spotify recommendations → learn from global music habits
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DATA ADVANTAGE (EXAMPLES)

- TikTok algorithm unbeatable due to engagement data
- YouTube's recommendation model trained on millions of hours
- OpenAI models improve as more developers use the API
- Instagram identifies trending content using engagement patterns

Real fact

- TikTok collects 10x more engagement signals than Instagram (swipe speed, watch time, replays).
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ZERO MARGINAL COST (DEFINITION)

- **“Cost to serve one more customer = almost zero.”**
- Once software is built → infinite copies
- No manufacturing required
- No shipping required

Examples

- Airbnb hosting new guest = no physical cost
 - ChatGPT serving 1 more user = negligible cost
 - Zoom adding another meeting participant = almost free
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ZERO MARGINAL COST (EXAMPLES)

- App Store apps → download costs nothing
- Spotify streams → cost per extra stream is tiny
- Netflix → unlimited viewers per movie

Impact

- Big platforms can offer products cheaply or free → impossible for small startups to compete.
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ENTRY BARRIERS IN DIGITAL MARKETS

- Huge server & talent cost
- Difficult to replicate data
- Hard to break user habits
- App ecosystems create dependence
- Regulatory compliance cost is high

Examples

- No new company can easily build a Google Search alternative
 - No competitor can instantly recreate Amazon's logistics
 - WhatsApp has 2B users → impossible to match overnight
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WINNER-TAKE-MOST MARKETS

- Digital markets rarely have “pure monopolies”

Rules

- Market leader gets most users
- A small competitor survives
- Others die out

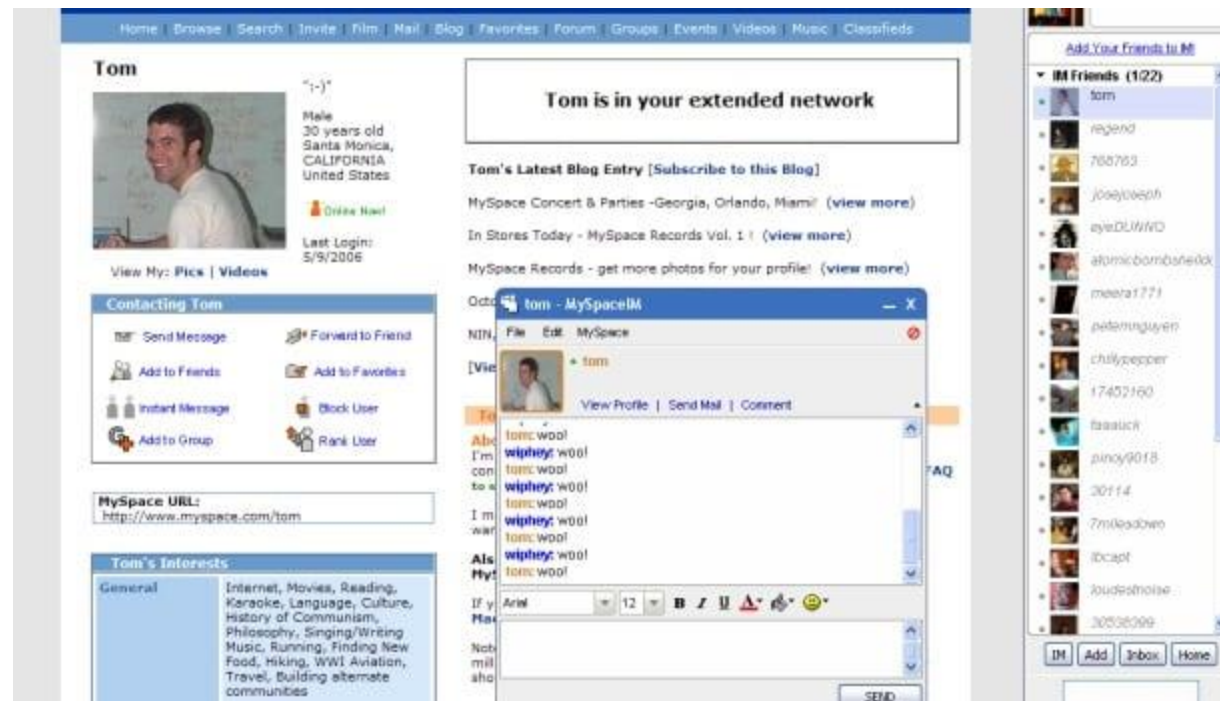
Examples

- Search: Google dominates, Bing survives
- Social media: Instagram + TikTok dominate, others fade
- E-commerce: Amazon leads, but smaller players exist

WHY WINNER-TAKE-MOST HAPPENS

- Strong network effects
- High switching costs
- Better data → better product
- Economies of scale force small players out
- Examples
 - WhatsApp + Viber in Nepal → WhatsApp won “most”
 - TikTok’s rapid rise drove Vine out
 - Instagram Stories killed Snapchat’s growth

MYSPEACE – PIONEER SOCIAL NETWORK DIED OUT



PLATFORM BUNDLING & LOCK-IN

- Big companies bundle services to keep users inside.
- Examples
 - Apple: iMessage + AirDrop + iCloud + FaceTime
 - Amazon: Prime Video + free delivery + Kindle + Music
 - Google: Maps + Search + Gmail all tied to one login
- Impact
 - Users rely on ecosystem → less likely to switch
 - Competitors must match entire bundle, not just one feature

“TIPPING POINTS” IN DIGITAL MARKETS

- **“The moment when one platform becomes unstoppable.”**
- Examples
 - WhatsApp hit 400M users → all friends already there → no reason to switch
 - TikTok reached 1B downloads → unbeatable algorithm
 - Amazon reached millions of sellers → customers stop checking competitors
- Real-world analogy
- Once a party becomes crowded → everyone wants to go there.

RISKS OF DIGITAL MONOPOLIES

- Less competition → lower innovation
- Privacy risk (data concentration)
- Price control (Apple 30% App Store tax)
- Biased algorithms
- Platforms can remove competition (Meta copying Stories, Reels, etc.)
- Examples
 - Google fined by EU for anti-competitive search practices
 - Apple sued for controlling iPhone app market
 - Amazon accused of undercutting sellers

COMPETITION POLICY CHALLENGES

- Governments struggle because:
- Products are “free” → hard to prove harm
- Digital markets move too fast
- Data ownership laws unclear

Examples

- TikTok banned in India for data concerns
- EU forcing Meta to allow cross-app messaging
- US investigating Amazon marketplace practices

GOVERNMENT TOOLS TO REGULATE BIG TECH

- **Data portability** (download your data)
- **Interoperability** (apps must work with each other)
- **Fines** for monopoly behavior
- **Breaking up companies** (rare but possible)
- Examples
 - EU GDPR → users can take their data
 - Digital Markets Act → forces Apple/Meta interoperability
 - US FTC cases against Amazon, Google

CASE STUDIES (OVERVIEW)

- Google Search → data + scale + zero marginal cost
- Meta → network effects + walled garden
- Amazon → economies of scale + logistics
- Apple → ecosystem lock-in + scope

SUMMARY SLIDE

- Key Forces Behind Digital Monopolies
- Network effects
- Economies of scale
- Economies of scope
- Data advantage
- Low marginal cost
- High switching costs

CASE STUDY: GOOGLE

- Best data
- Largest index of websites
- Superior algorithm
- Default browser integration (Chrome/Android)
- Ads funding expansion
- Result
- Competitors like Yahoo, Bing, DuckDuckGo cannot catch up

CASE STUDY: AMAZON

- Why Amazon dominates
- Gigantic seller ecosystem
- Fast & cheap delivery
- Prime lock-in
- Uses data to optimize delivery & pricing
- Result
- Difficult for smaller e-commerce sites to compete